

Trigonometric Identities — Solutions

Warm-up

Question 1:

$$\begin{aligned}\tan(A - B) &= \frac{\tan A - \tan B}{1 + \tan A \tan B} \\ \frac{\tan 5\theta - \tan 2\theta}{1 + \tan 5\theta \tan 2\theta} &= \tan(5\theta - 2\theta) \\ &= \tan(3\theta)\end{aligned}$$

Question 2:

$$\begin{aligned}\sin(A + B) &= \sin A \cos B + \cos A \sin B \\ \sin(2x + x) &= \sin 2x \cos x + \cos 2x \sin x\end{aligned}$$

Question 3:

$$\begin{aligned}\sin^2 \theta &= 1 - \left(\frac{3}{5}\right)^2 = \frac{16}{25} \\ \sin \theta &= \frac{4}{5} \\ \sin 2\theta &= 2 \sin \theta \cos \theta \\ &= 2 \left(\frac{4}{5}\right) \left(\frac{3}{5}\right) \\ &= \frac{24}{25}\end{aligned}$$

Question 4:

$$\begin{aligned}\cos 2A &= \cos^2 A - \sin^2 A \\ \cos 2A &= 2 \cos^2 A - 1 \\ \cos 2A &= 1 - 2 \sin^2 A\end{aligned}$$

Question 5:

$$\begin{aligned}\sin 2\theta &= 2 \sin \theta \cos \theta \\ 2 \sin 15^\circ \cos 15^\circ &= \sin(2 \times 15^\circ) \\ &= \sin 30^\circ \\ &= \frac{1}{2}\end{aligned}$$

Question 6:

$$\begin{aligned}\tan 2A &= \frac{2 \tan A}{1 - \tan^2 A} \\ \tan 2A &= \frac{2(2)}{1 - (2)^2} \\ &= \frac{4}{-3} \\ &= -\frac{4}{3}\end{aligned}$$

Question 7:

$$\begin{aligned}\cos 2\theta &= 1 - 2\sin^2 \theta \\ 1 - 2\sin^2(3x) &= \cos(2 \times 3x) \\ &= \cos 6x\end{aligned}$$

Question 8:

$$\frac{2\sin \theta \cos \theta}{2\sin \theta} = \cos \theta$$

Question 9:

$$\cos^2 \alpha - \sin^2 \alpha = \cos 2\alpha$$

Question 10:

$$\begin{aligned}\sin 15^\circ &= \sin(45^\circ - 30^\circ) \\ &= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ \\ &= \left(\frac{\sqrt{2}}{2}\right) \left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right) \left(\frac{1}{2}\right) \\ &= \frac{\sqrt{6} - \sqrt{2}}{4}\end{aligned}$$

Skill-building

Question 1:

$$\begin{aligned}\frac{\sin 2x}{1 + \cos 2x} &= \frac{2\sin x \cos x}{1 + (2\cos^2 x - 1)} \\ &= \frac{2\sin x \cos x}{2\cos^2 x} \\ &= \frac{\sin x}{\cos x} \\ &= \tan x\end{aligned}$$

Question 2:

$$\begin{aligned}\cos A - \cos 3A &= 2\sin 2A \sin A \\ \sin 3A - \sin A &= 2\cos 2A \sin A \\ \frac{2\sin 2A \sin A}{2\cos 2A \sin A} &= \frac{\sin 2A}{\cos 2A} \\ &= \tan 2A\end{aligned}$$

Question 3:

$$\begin{aligned}\sin 2\theta &= \frac{1}{2} \\ 2\theta &= \frac{\pi}{6}, \frac{5\pi}{6} \\ \theta &= \frac{\pi}{12}, \frac{5\pi}{12}\end{aligned}$$

Question 4:

$$\begin{aligned}\frac{7}{25} &= 2 \cos^2 \theta - 1 \\ 2 \cos^2 \theta &= \frac{32}{25} \\ \cos^2 \theta &= \frac{16}{25} \\ \cos \theta &= -\frac{4}{5}\end{aligned}$$

Question 5:

$$\begin{aligned}\cos^4 \theta - \sin^4 \theta &= (\cos^2 \theta - \sin^2 \theta)(\cos^2 \theta + \sin^2 \theta) \\ &= (\cos^2 \theta - \sin^2 \theta)(1) \\ &= \cos 2\theta\end{aligned}$$

Question 6:

$$\begin{aligned}\frac{\sin 3\alpha \cos \alpha - \cos 3\alpha \sin \alpha}{\sin \alpha \cos \alpha} &= \frac{\sin(3\alpha - \alpha)}{\sin \alpha \cos \alpha} \\ &= \frac{\sin 2\alpha}{\sin \alpha \cos \alpha} \\ &= \frac{2 \sin \alpha \cos \alpha}{\sin \alpha \cos \alpha} \\ &= 2\end{aligned}$$

Question 7:

$$\begin{aligned}(2 \cos^2 x - 1) + 3 \cos x + 2 &= 0 \\ 2 \cos^2 x + 3 \cos x + 1 &= 0 \\ (2 \cos x + 1)(\cos x + 1) &= 0 \\ \cos x &= -\frac{1}{2}, -1 \\ x &= \frac{2\pi}{3}, \pi, \frac{4\pi}{3}\end{aligned}$$

Question 8:

$$\begin{aligned}\frac{1}{\sin 2\theta} + \frac{\cos 2\theta}{\sin 2\theta} &= \frac{1 + \cos 2\theta}{\sin 2\theta} \\ &= \frac{2 \cos^2 \theta}{2 \sin \theta \cos \theta} \\ &= \frac{\cos \theta}{\sin \theta} \\ &= \cot \theta\end{aligned}$$

Question 9:

$$\begin{aligned}\sin 2x &= \frac{2 \tan x}{1 + \tan^2 x} \\ &= \frac{2t}{1 + t^2}\end{aligned}$$

Question 10:

$$\begin{aligned}\sin 5x \cos 3x &= \frac{1}{2}(\sin(5x + 3x) + \sin(5x - 3x)) \\ &= \frac{1}{2}(\sin 8x + \sin 2x)\end{aligned}$$

Easier Exam Questions

Question 1:

$$\begin{aligned}\frac{\tan 3x - \tan x}{1 + \tan 3x \tan x} &= \tan(3x - x) \\ &= \tan 2x \quad (\text{Option A})\end{aligned}$$

Question 2:

$$\begin{aligned}\frac{1 + \tan 5x \tan 3x}{\tan 5x - \tan 3x} &= \frac{1}{\tan(5x - 3x)} \\ &= \cot 2x \quad (\text{Option A})\end{aligned}$$

Question 3:

$$\begin{aligned}\frac{1 - \cos 2x}{\sin 2x} &= \frac{2 \sin^2 x}{2 \sin x \cos x} \\ &= \tan x \quad (\text{Option D})\end{aligned}$$

Question 4:

$$\begin{aligned}\cos 2\theta &= 1 - 2 \sin^2 \theta \\ &= 1 - 2 \left(\frac{12}{13} \right)^2 \\ &= 1 - \frac{288}{169} \\ &= -\frac{119}{169} \quad (\text{Option D})\end{aligned}$$

Question 5:

$$\begin{aligned}\frac{\cos 3x - \cos 5x}{\sin 3x + \sin 5x} &= \frac{2 \sin 4x \sin x}{2 \sin 4x \cos x} \\ &= \tan x\end{aligned}$$

Question 6:

$$\begin{aligned}\frac{1 - \cos 2x + \sin 2x}{1 + \cos 2x + \sin 2x} &= \frac{2 \sin^2 x + 2 \sin x \cos x}{2 \cos^2 x + 2 \sin x \cos x} \\ &= \frac{2 \sin x (\sin x + \cos x)}{2 \cos x (\cos x + \sin x)} \\ &= \tan x\end{aligned}$$

Harder Exam Questions

Question 1:

$$\begin{aligned}2u^2 - u - 1 &= 0 \\ (2u + 1)(u - 1) &= 0 \\ \cos 2x &= -\frac{1}{2}, 1 \\ x &= 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi\end{aligned}$$

Question 2:

$$\begin{aligned}\text{LHS} &= \tan \theta + \tan^2 \theta + 1 + \cot \theta \\ &= (\tan \theta + \cot \theta) + \sec^2 \theta \\ &= \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} + \frac{1}{\cos^2 \theta} \\ &= \frac{\cos \theta + \sin \theta}{\sin \theta \cos^2 \theta} \\ \text{RHS} &= \frac{\tan \theta + 1}{\sin \theta \cos \theta} = \frac{\sin \theta + \cos \theta}{\sin \theta \cos^2 \theta}\end{aligned}$$

Question 3(a):

$$\begin{aligned}\cos 105^\circ &= \frac{\sqrt{2} - \sqrt{6}}{4} \\ \sec 105^\circ &= \frac{4}{\sqrt{2} - \sqrt{6}} \\ &= -(\sqrt{2} + \sqrt{6})\end{aligned}$$

Question 3(b)i:

$$\begin{aligned}\sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}} &= \sqrt{\frac{2 \sin^2 \theta}{2 \cos^2 \theta}} \\ &= \tan \theta\end{aligned}$$

Question 3(b)ii:

$$\begin{aligned}\tan 22.5^\circ &= \sqrt{\frac{1 - \cos 45^\circ}{1 + \cos 45^\circ}} \\ &= \sqrt{\frac{2 - \sqrt{2}}{2 + \sqrt{2}}} \\ &= \sqrt{2} - 1\end{aligned}$$

Question 3(d):

$$\begin{aligned}\sin(A - 2B) \cos(A + 2B) &= \frac{1}{2}(\sin(2A) + \sin(-4B)) \\ &= \frac{1}{2}(\sin 2A - \sin 4B)\end{aligned}$$

Question 3(e):

$$\begin{aligned}\frac{2(\sin A + \cos A)(\sin^2 A - \sin A \cos A + \cos^2 A)}{\sin A + \cos A} &= 2(1 - \sin A \cos A) \\ &= 2 - \sin 2A\end{aligned}$$

Question 4(i):

$$\begin{aligned}\cot \theta - \cot 2\theta &= \frac{\cos \theta}{\sin \theta} - \frac{\cos 2\theta}{\sin 2\theta} \\ &= \frac{\sin(2\theta - \theta)}{\sin \theta \sin 2\theta} \\ &= \csc 2\theta\end{aligned}$$

Question 4(ii):

$$\begin{aligned}\cot 15^\circ &= \csc 30^\circ + \cot 30^\circ \\ &= 2 + \sqrt{3}\end{aligned}$$

Question 5:

$$\begin{aligned}\tan\left(\frac{\pi}{4} + \theta\right) + \tan\left(\frac{\pi}{4} - \theta\right) &= \frac{2(1 + \tan^2 \theta)}{1 - \tan^2 \theta} \\ &= \frac{2 \sec^2 \theta}{1 - \tan^2 \theta} \\ &= \frac{2}{\cos 2\theta} \\ &= 2 \sec 2\theta\end{aligned}$$

Question 6(i):

$$\sin mx \sin \frac{x}{2} = \frac{1}{2}(\cos(m - \frac{1}{2})x - \cos(m + \frac{1}{2})x)$$

Question 6(ii):

$$\begin{aligned}2 \sin \frac{x}{2} \sum \sin kx &= \sum (\cos(k - \frac{1}{2})x - \cos(k + \frac{1}{2})x) \\ &= \cos \frac{x}{2} - \cos(n + \frac{1}{2})x \\ \sum \sin kx &= \frac{\cos \frac{x}{2} - \cos(n + \frac{1}{2})x}{2 \sin \frac{x}{2}}\end{aligned}$$

Question 7(i):

$$\begin{aligned}(\sin \alpha - \cos \alpha)^2 &= \sin^2 \alpha - 2 \sin \alpha \cos \alpha + \cos^2 \alpha \\ &= 1 - \sin 2\alpha\end{aligned}$$

Question 7(ii):

$$\begin{aligned}y^2 &= 1 - \sin 150^\circ = \frac{1}{2} \\ y &= \frac{\sqrt{2}}{2}\end{aligned}$$

Question 8(a):

$$\begin{aligned}(\sin 22.5^\circ + \cos 22.5^\circ)^2 &= 1 + \sin 45^\circ \\ &= \frac{2 + \sqrt{2}}{2}\end{aligned}$$

Question 8(b):

$$\begin{aligned}\tan x &= -\frac{1}{2\sqrt{2}}, \quad \tan y = \sqrt{2} \\ \tan(x + y) &= \frac{-\frac{1}{2\sqrt{2}} + \sqrt{2}}{1 - (-\frac{1}{2\sqrt{2}})(\sqrt{2})} \\ &= \frac{\sqrt{2}}{2}\end{aligned}$$

Question 8(c):

$$\begin{aligned}\frac{\sin 3\theta \cos \theta + \cos 3\theta \sin \theta}{\sin \theta \cos \theta} &= \frac{\sin 4\theta}{\sin \theta \cos \theta} \\ &= \frac{2 \sin 2\theta \cos 2\theta}{\sin \theta \cos \theta} \\ &= 4 \cos 2\theta\end{aligned}$$

Question 9:

$$\begin{aligned}\cos 2x + \cos 4x + \cos 6x &= -1 \\ 2 \cos 2x(2 \cos 2x - 1)(\cos 2x + 1) &= 0 \\ x &= \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}, \frac{11\pi}{6}\end{aligned}$$